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Transportation unit for a conveyor

Abstract

A transportation unit for a conveyor includes a first shuttle and a second shuttle arranged to move along a planar conveying track independently of each other; and a gripping device including a first jaw mounted so that it can rotate on the first shuttle about a first rotational axis and a second jaw mounted so that it can rotate on the second shuttle about a second rotational axis parallel to the first rotational axis. The first jaw and the second jaw are interconnected by a sliding connection along a translation axis orthogonal to the first and second rotational axes to form the clamping jaws of a clamp having a clamping axis parallel to the translation axis.

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Background/Summary

DESCRIPTION

[0001] The invention relates to the conveying of receptacles within an industrial production line, more particularly to a transportation unit for a conveyor.

BACKGROUND OF THE INVENTION

[0002] It is known to use a conveyor for transporting empty bottles from a blowing machine to a filling machine within an industrial production line. Conveyors generally comprise a conveying and a track plurality of transportation units which are arranged to move along said conveying track by means of electromagnetic currents.

[0003] Transportation units comprising a first shuttle and a second shuttle arranged to move along the conveying track independently of each other are known. The first shuttle and the second shuttle bear a first jaw and a second jaw, respectively, which form the clamping jaws of a clamp.

[0004] The first jaw and the second jaw are fixed in place with respect to the first shuttle and the second shuttle, respectively, and are arranged so that bringing said first and second shuttles together allows a bottle to be grasped and so that moving said shuttles in synchronisation allows said bottle to be moved along the conveying track.

[0005] The conveying track generally comprises a sequence of straight portions and curved portions to form a closed loop. When the transportation unit is moving to bring the bottle from the blowing machine to the filling machine, passing from a straight portion to a curved portion causes the first jaw to rotate with respect to the second jaw and therefore causes the bottle to slide between the first and second jaws, which can cause said bottle to fall and thus break.

OBJECT OF THE INVENTION

[0006] The object of the invention is to propose a transportation unit for a conveyor for overcoming the above-mentioned drawback at least in part.

SUMMARY OF THE INVENTION

[0007] To this end, a transportation unit for a conveyor is proposed, comprising: [0008] a first shuttle and a second shuttle arranged to move along a planar conveying track independently of each other; and [0009] a gripping device comprising a first jaw mounted so that it can rotate on the first shuttle about a first rotational axis and a second jaw mounted so that it can rotate on the second shuttle about a second rotational axis parallel to the first rotational axis.

[0010] According to the invention, the first jaw and the second jaw are interconnected by a sliding connection along a translation axis orthogonal to the first and second rotational axes to form the clamping jaws of a clamp having a clamping axis parallel to the translation axis.

[0011] The sliding connection between the first jaw and the second jaw makes it possible to hold said first jaw and said second jaw axially facing each other, regardless of the angular position of the first and second shuttles along the conveying track.

[0012] According to a particular feature, the clamping axis of the first and second jaws is perpendicular to the first rotational axis and the second rotational axis of the gripping device.

[0013] According to a particular feature, the sliding connection comprises a first arm rigidly connected to the first jaw, a second arm rigidly connected to the second jaw, and a rod having a first end rigidly attached to either the first or the second arm and a second end mounted so that it can move in translation in a hole formed in the other of the first and the second arm.

[0014] In particular, the first arm and the second arm are identical.

[0015] According to another particular feature, at least one out of the first jaw and the second jaw comprises a centring V.

[0016] According to another particular feature, at least one out of the first jaw and the second jaw is flexible.

[0017] According to another particular feature, the first shuttle and the second shuttle are identical.

[0018] The invention also relates to a conveyor comprising at least one such transportation unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] The invention will be better understood in the light of the following description, which is purely illustrative and non-limiting and should be read with reference to the accompanying drawings, in which:

[0020] FIG. **1** is a perspective view showing a conveyor comprising a transportation unit according to a particular embodiment of the invention;

[0021] FIG. **2**A is a first perspective view of the transportation unit shown in FIG. **1**;

[0022] FIG. **2**B is a second perspective view of the transportation unit shown in FIG. **1**;

[0023] FIG. **2**C is an identical view to FIG. **2**A in which the transportation unit is carrying a receptacle;

[0024] FIG. **3**A is a plan view of the transportation unit shown in FIG. **1**, arranged on a straight portion of the conveying track;

[0025] FIG. **3**B is an identical view to FIG. **3**A in which the transportation unit is carrying a receptacle;

[0026] FIG. **3**C is an identical view to FIG. **3**A, arranged on a curved portion of the conveying track;

[0027] FIG. **4** is a plan view of a variant of the transportation unit shown in FIG. **1**.

DETAILED DESCRIPTION OF THE INVENTION

[0028] With reference to FIG. **1**, a receptacle conveyor **1** according to a particular embodiment of the invention comprises a conveying track **2** and a plurality of transportation units **10** capable of moving along said conveying track **2** by means of electromagnetic currents. For reasons of clarity, only one transportation unit **10** has been shown but the transportation units **10** are all identical here.

[0029] The conveying track **2** comprises an outer periphery **3** defining a height of the conveying track **2**. The outer periphery **3** of the conveying track **2** extends substantially vertically and comprises two parallel planar surfaces **3**.**1** interconnected by two semi-cylindrical surfaces **3**.**2** so as to define a movement path of the transportation unit **10** in a closed loop.

[0030] With reference to FIGS. **2**A-**2**C, the transportation unit **10** comprises a first shuttle **100**.**1**, a second shuttle **100**.**2** and a gripping device **200** borne by the first shuttle **100**.**1** and the second shuttle **100**.**2**.

[0031] In this case, the first shuttle **100**.**1** and the second shuttle **100**.**2** are identical and are arranged to move horizontally, independently of each other, along the outer periphery **3** of the conveying track **2**.

[0032] The first and second shuttles **100**.**1**, **100**.**2** each comprise a body **101** extending substantially vertically, facing the outer periphery **3** of the conveying track **2**. The body **101** includes an upper part bearing two first castors **102**, and a lower part bearing two second castors **103**.

[0033] The first castors **102** are mounted on the body **101** so that they can rotate about first vertical axes Z.sub.102 and are shaped to roll in grooves **4**.**1**, **4**.**2** formed in the upper part of the outer periphery **3** of the conveying track **2**. The grooves **4**.**1**, **4**.**2** each extend in a first horizontal plane and define a planar movement path for the first and second shuttles **100**.**1**, **100**.**2**.

[0034] The second castors **103** are mounted on the body **101** so that they can rotate about second vertical axes Z.sub.103 and are shaped to roll on a planar rolling surface **4**.**3** formed in the lower part of the outer periphery **3** of the conveying track **2**. The rolling surface **4**.**3** extends in a second horizontal plane, and in this case the second rotational axes Z.sub.103 of the second castors **103** are coaxial with the first rotational axes Z.sub.102 of the first castors **102**.

[0035] It is understood that the first shuttle **100**.**1** and the second shuttle **100**.**2** are arranged to roll on the outer periphery **3** of the conveying track **2**.

[0036] The first shuttle **100**.**1** and the second shuttle **100**.**2** are held and moved, independently of

each other, along the outer periphery **3** of the conveying track **2** by means of an electromagnetic system that is known per se; the position, speed and acceleration of the first shuttle **100.1** are managed independently of the position, speed and acceleration of the second shuttle **100.2**. **25**
[0037] The first and second shuttles **100.1**, **100.2** also each include a substantially rectilinear support **104** that extends generally horizontally from an upper end of the body **101**. The support **104** has a proximal end rigidly attached to the upper end of the body **101**, and a distal end having a hole in which a hinge shaft **201** is mounted so that it can rotate about a vertical axis Z.sub.201. It is understood that the support **104** is stationary with respect to the body **101**. In this case, the hinge shaft **201** has a lower end and an upper end extending on either side of the support **104**.
[0038] The gripping device **200** forms a clamp and comprises: [0039] the hinge shafts **201** borne by the first shuttle **100.1** and the second shuttle **100.2**; [0040] a first jaw **210** rigidly attached to the lower end of the hinge shaft **201** borne by the first shuttle **100.1**; [0041] a second jaw **220** rigidly attached to the lower end of the hinge shaft **201** borne by the second shuttle **100.2**; [0042] a first arm **202.1** having a proximal end rigidly attached to the upper end of the hinge shaft **201** borne by the first shuttle **100.1**; [0043] a second arm **202.2** having a proximal end rigidly attached to the upper end of the hinge shaft **201** borne by the second shuttle **100.2**; and [0044] a holding rod **203** facing the first and second jaws **110**, **120**, the rod **103** having a first end rigidly attached to a distal end of the second arm **202.2**, and a second end, on the opposite side to the first end, mounted so that it can move in translation in a hole formed at a distal end of the first arm **202.1**, along a longitudinal axis X**203** of the rod **203**.
[0045] It should be noted that the longitudinal axis X.sub.203 of the rod **203** is orthogonal to the rotational axes Z.sub.201 of the hinge shafts **201**.
[0046] It is understood that the first jaw **210** and the first arm **202.1** are stationary with respect to the hinge shaft **201** borne by the first shuttle **100.1** (in other words that the first jaw **210** is stationary with respect to the first arm **202.1**), and that the second jaw **220**, the second arm **202.2** and the rod **203** are stationary with respect to the hinge shaft **201** borne by the second shuttle **100.2** (in other words that the second jaw **220** is stationary with respect to the second arm **202.2**).
[0047] It is also understood that the first jaw **210** and the second jaw **220** form the clamping jaws of the clamp and can move in translation along the longitudinal axis X.sub.203 of the rod **203** which defines a sliding connection.
[0048] It is also understood that the first jaw **210** cannot rotate relative to the second jaw **120**.
[0049] The first jaw **210** projects horizontally from the hinge shaft **201** borne by the first shuttle **100.1**, along a first clamping axis X.sub.210 that is perpendicular to the rotational axis Z.sub.201 of said hinge shaft **201** and parallel to the longitudinal axis X.sub.203 of the rod **203** (FIG. **3**A). It is understood that the first clamping axis X.sub.210 and the rotational axis Z.sub.201 of the hinge shaft **201** borne by the first shuttle **100.1** intersect.
[0050] The first jaw **210** comprises a free end that faces the second jaw **220** and that is arranged to form a first centring V **211.1** and a second centring V **211.2** arranged one above the other. The first centring V **211.1** and the second centring V **211.2** extend, respectively, in a first horizontal plane defining a first clamping height H.sub.1 of the receptacle R and in a second horizontal plane defining a second clamping height H.sub.2 of the receptacle R (FIG. **2**C). In this case, the first centring V **211.1** and the second centring V **211.2** are identical and have the same vertical plane of symmetry, defined here by the rotational axes Z.sub.201 of the hinge shafts **201**. The shape and dimensions of the first and second centring Vs **211.1**, **211.2** are adapted to the shape and dimensions of the receptacle R to be transported.
[0051] The first centring V **211.1** and the second centring V **211.2** are referred to as being rigid in that they are not deformed when the receptacle R is clamped as set out below.
[0052] The second jaw **220** projects horizontally from the hinge shaft **201** borne by the second shuttle **100.2**, along a second clamping axis X.sub.220 that is perpendicular to the rotational axis Z.sub.201 of said hinge shaft **201** and to the longitudinal axis X.sub.203 of the rod **203** (FIG. **3**A).

In this case, the second clamping axis X.sub.120 is coaxial with the first clamping axis X.sub.110. It is understood that the second clamping axis X.sub.220 and the rotational axis Z.sub.201 of the hinge shaft **201** borne by the second shuttle **100.2** intersect.

[0053] The second jaw **220** comprises a free end that faces the first jaw **210** and that is arranged to form a first clamping flange **221.1** and a second clamping flange **221.2** arranged one above the other. The first clamping flange **221.1** and the second clamping flange **221.2** extend, respectively, in the first horizontal plane defining the first clamping height H.sub.1 of the receptacle R and in the second horizontal plane defining the second clamping height H.sub.2 of the receptacle R. In this case, the first clamping flange **221.1** and the second clamping flange **221.2** are identical and have the same vertical plane of symmetry, defined here, as with the first and second centring Vs **211.1**, **211.2**, by the rotational axes Z.sub.201 of the hinge shafts **201**. With reference to FIG. **3**A, the first and second clamping flanges **221.1**, **221.2** each comprise a stationary base **222**, generally in the shape of a circular arc, through the middle of which there extends a rod **223** for controlling the deformation of the first and second clamping flanges **221.1**, **211.2**. The rod **223** has a longitudinal axis X.sub.223 that is coaxial with the clamping axis X.sub.220 of the second jaw **220**, and is mounted so that it can move in translation in the base **222** along its longitudinal axis X.sub.223 between an extended position, in which a distal end of the rod **223** is remote from the base **222**, and a retracted position, in which the distal end of the rod **223** is close to said base **222**.

[0054] The distal end of the rod **223** is coupled to a first end **222.1** and to a second end **222.2** of the base **222** by means of a first link chain **224.1** and a second link chain **224.2**, respectively. In this case, the first link chain **224.1** and the second link chain **224.2** are identical and comprise three interconnected links, namely a first link **224***a*, a second link **224***b* and a third link **224***c*. The first link **224***a* is coupled to the base **224** along a first vertical axis, the second link **224***b* is coupled to the first link **224***a* along a second vertical axis, and the third link **224***c* is coupled to the second link **224***b* along a third vertical axis and to the distal end of the rod **223** along a fourth vertical axis. A free end of the first link **224***a* of the first and second link chains **224.1**, **224.2** is also coupled, by means of a helical tension spring **225**, to a free end of an attachment tab **226.1**, **226.2** projecting radially outwards from the base **222**.

[0055] It is understood that the links **224***a*, **224***b*, **224***c* of the first link chain **224.1** form, together with the base **222** and the rod **223**, a first horizontal loop, and that the links **224***a*, **224***b*, **224***c* of the second link chain **224.2** form, together with the base **222** and the rod **223**, a second horizontal loop identical to the first loop. It is also understood that: [0056] when the rod **223** is in the extended position, the links **224***a*, **224***b*, **224***c* of the first link chain **224.1** are remote from the links **224***a*, **224***b*, **224***c* of the second link chain **224.2**; and [0057] when the rod **223** is in the retracted position, the links **224***a*, **224***b*, **224***c* of the first link chain **224.1** are remote from the links **224***a*, **224***b*, **224***c* of the second link chain **224.2**; and [0058] the rod **223** is returned to the extended position by the springs **225**.

[0059] In this case, the dimensions of the links **224***a*, **224***b*, **224***c* of the first and second link chains **224.1**, **224.2** are selected such that: [0060] when the rod **223** is in the extended position, the second and third links **123***b*, **123***c* of the first and second link chains **123.1**, **123.2** are substantially horizontally aligned along an axis that is perpendicular to the longitudinal axis X**223** of the rod **223** and therefore perpendicular to the clamping axis X**220**; and [0061] when the rod **223** is in the retracted position, the second and third links **224***b*, **224***c* of the first and second link chains **224.1**, **224.2** form a recess suitable for matching the shape of the receptacle R to be transported.

[0062] It should be noted that the movement of the rod **223** is not controlled by an actuator; the movement of the rod **223** results solely from a sufficient axial force exerted on the distal end of said rod **223**.

[0063] The first clamping flange **221.1** and the second clamping flange **221.2** are referred to as being deformable or flexible in that the links **224***a*, **224***b*, **224***c* of the first and second link chains **224.1**, **224.2** are made to move when the receptacle R is clamped as set out below.

[0064] The operation of the transportation unit **10** and its gripping device **200** will now be described in detail.

[0065] In this case, the receptacle R intended to be transported is a bottle having, in a manner known per se, a lower part of a substantially constant diameter, an upper part that narrows upwards and a neck that forms an upper end of the bottle.

[0066] First, the first shuttle **100.1** and the second shuttle **100.2** of the transportation unit **10** are locked along the conveying track **2** so as to be far enough apart from each other to accommodate the bottle R in a vertical position between the first jaw **210** and the second jaw **220**.

[0067] Second, the bottle R is fed in and held in a vertical position, for example by means of a transfer robot, between the first jaw **210** and the second jaw **220** such that the longitudinal axis of the bottle R is substantially perpendicular to the first and second clamping axes X.sub.210, X.sub.220 of said first and second jaws **210**, **220**. It is understood that the longitudinal axis of the bottle R intersect the first and second clamping axes X.sub.210, X.sub.220 of the first and second jaws **210**, **220**.

[0068] The first shuttle **100.1** and the second shuttle **100.2** are then moved towards each other so that the distance between the first jaw **110** and the second jaw decreases until said first jaw **210** and said second jaw **220** are in contact with a first lateral surface and a second lateral surface of the bottle R at a first height H.sub.1 and a second height H.sub.2 of said bottle R, respectively. The bottle R is then accommodated in the centring Vs **211.1**, **211.2** of the first jaw **210**.

[0069] As the second shuttle **100.2** is brought ever closer to the first shuttle **100.1**, the bottle R exerts a compression force on the sliding rods **223** of the second jaw **220**, said force urging said rods **223** from the extended position to the retracted position. The second and third links **224***b*, **224***c* of the first and second link chains **224.1**, **224.2** then substantially match the shape of the second lateral surface of the bottle R.

[0070] The second shuttle **100.2** is moved further towards the first shuttle **100.1** until the clamping forces exerted by the first and second jaws **210**, **220** on the lateral surfaces of the bottle R are sufficient to hold said bottle R in position between the first and second jaws **210**, **220** without the help of the transfer robot.

[0071] The bottle R is then released by the transfer robot, and the movements of the first shuttle **100.1** and second shuttle **100.2** are controlled and synchronised so that both the space between them and the clamping forces exerted on the bottle R by the first and second jaws **210**, **220** remain substantially constant throughout the movement of the first and second shuttles **100.1**, **100.2** along the conveying track **2**. The bottle R is then made to move along the conveying track **2**.

[0072] To release the bottle R from the clamping forces exerted by the first and second jaws **210**, **220** in order to remove it from the conveyor **1**, the movements of the first and second shuttles **100.1**, **100.2** merely have to be controlled so as to move the shuttles away from each other.

[0073] It should be noted that the sliding connection formed by the rod **203** between the first jaw **210** and the second jaw **220** makes it possible to hold said first jaw **210** and said second jaw **220** axially facing each other, regardless of the angular position of the first and second shuttles **100.1**, **100.2**.

[0074] Thus, regardless of whether the first shuttle **100.1** and/or the second shuttle **100.2** are rolling on the planar surfaces **3.1** (FIGS. **3**A-**3**B) or the semi-cylindrical surfaces **3.2** (FIG. **3**C) of the conveying track **2**, the contact regions between the bottle R and the first and second jaws **210**, **220** remain identical throughout the movement of the transportation unit **10** along the conveying track **2**, such that the risk of the bottle R falling while it is being transported is particularly limited.

[0075] FIG. **4** shows a transportation unit **10′** that is merely a variant of the transportation unit **10**. The transportation unit **10′** differs from the transportation unit **10** in that the gripping device **200′** of the transportation unit **10′** comprises a first jaw **210′** and a second jaw **220′** identical to the first jaw **210′**. Both the jaw **110′** and the second jaw **120′** of the first transportation unit **10′** are identical to the second, deformable jaw **220** of the transportation unit **10**.

[0076] The transportation unit **10′** is operated in the same way as the transportation unit **10**, except that the bottle R is clamped by the first jaw **210′** and the second jaw **220′** being deformed. The sliding connection formed by the rod **203** between the first jaw **210′** and the second jaw **220′** makes it possible, as in the transportation unit **10**, to hold the first jaw **210′** and the second jaw **220′** axially facing each other, regardless of the angular position of the first and second shuttles **100.1**, **100.2**.
[0077] Another variant of the transportation unit **10** (not shown) involves replacing the second, deformable jaw **220** with a jaw identical to the first, rigid jaw **210**.
[0078] This other variant of the transportation unit **10** is operated in the same way as the transportation unit **10**, except that the bottle R is clamped without the first and second jaws being deformed. The sliding connection formed by the rod **203** between the first jaw and the second jaw makes it possible, as in the transportation unit **10**, to hold the first jaw and the second jaw axially facing each other, regardless of the angular position of the first and second shuttles **100.1**, **100.2**.
[0079] It goes without saying that the invention is not limited to the described embodiment but covers any variant falling under the scope of the invention.
[0080] Although in this case the first jaw **210** and the second jaw **220** comprise two centring Vs **211.1**, **211.2** and two clamping flanges **221.1**, **221.2** to form two clamping levels, the first jaw **210** and/or the second jaw **220** may comprise one or more than two centring Vs and one or more than two clamping flanges so as to form one or more than two clamping levels.
[0081] The springs **225** can be replaced with any resilient return means capable of returning the sliding rod **223** to its extended position. For example, a spring may be mounted around the rod between the distal end of the rod and the base.
[0082] The sliding rod **223** may be replaced with any means for ensuring a sliding connection between the first jaw **210**, **210′** and the second jaw **220**, **220′**.
[0083] The dimensions and shape of the different elements of the transportation unit **10**, **10′** can be adapted to the dimensions and shape of the receptacles R to be transported, in particular those of the arms **202.1**, **202.2**, of the centring Vs **211.1**, **211.2**, of the links **224***a*, **224***b*, **224***c*, etc.
[0084] Although in this case the link chains **224.1**, **224.2** comprise three links **224***a*, **224***b*, **224***c*, they may comprise a number of links that is less than or greater than three.
[0085] The first jaw **210** and the second jaw **220** can be attached to the first arm **202.1** and the second arm **202.2**, respectively.
[0086] The first arm **202.1** and the second arm **202.2** can be attached to the first jaw **210** and the second jaw **220**, respectively. The first shuttle **100.1** may be different from the second shuttle **100.2**.
[0087] The first arm **202.1** may be different from the second arm **202.2**.
[0088] Although in this case the clamping axis X.sub.210, X.sub.220 of the first and second jaws **210**, **210′**, **220**, **220′** is perpendicular to the rotational axes Z.sub.201 of the hinge shafts **201**, it may more generally be orthogonal to said rotational axes Z.sub.201.
[0089] Although in this case the centring Vs **211.1**, **211.2** of the first jaw **210** and the clamping flanges **221.1**, **221.2** of the second jaw **220** extend in a horizontal plane, they can extend in a plane that forms a non-zero angle with the horizontal plane so as to clamp a bottle R whose longitudinal axis is not vertical.
[0090] Although in this case the conveying track **2** extends in a horizontal plane, it may extend in a slanted plane that forms a non-zero angle with the horizontal plane.
[0091] Such slants of the conveying track **2** and of the first and second jaws **210**, **210′**, **220**, **220′** may in particular make it possible to have a different slant of the longitudinal axis of the bottle R along the planar surface **3.1** and also of the planar surface **3.2** of the conveying track **2**. For example, when the conveying track **2** is slanted by 45° about its longitudinal axis and the first and second jaws **210**, **210′**, **220**, **220′** are also slanted by 45° with respect to the plane in which the conveying track **2** extends, the longitudinal axis of the bottle R is horizontal on one of the planar surfaces **3.1**, **3.2** of the conveying track **2** and vertical on the other of the planar surfaces **3.1**, **3.2**.

[0092] The position of the sliding connection with respect to the first and second jaws **210**, **210′**, **220**, **220′** can be different from that shown. By way of example, the sliding connection may be arranged above or below the first and second jaws **210**, **210′**, **220**, **220′**.
[0093] To improve the hold of the bottle R by means of the first and second jaws **210**, **220**, the centring Vs **211.1**, **211.2** and the clamping flanges **221.1**, **221.2** may be covered, at least in part, with an elastomer material so as to provide contact surfaces having a high coefficient of friction with the bottle R. For example, an elastomer profile may be inserted on the inner periphery of the centring Vs **221.1**, **221.2**, and an elastomer pellet may be clipped onto an outer surface of the second links **224***b* of the first and second link chains **224.1**, **224.2**.
[0094] The dimensions and shape of the different elements forming the gripping device **200**, **200′** can be adapted to the dimensions and shape of different ranges of receptacles R to be transported.

Claims

1. A transportation unit for a conveyor, comprising: a first shuttle and a second shuttle arranged to move along a planar conveying track independently of each other; and a gripping device comprising a first jaw mounted so that it can rotate on the first shuttle about a first rotational axis and a second jaw mounted so that it can rotate on the second shuttle about a second rotational axis parallel to the first rotational axis, the first jaw and the second jaw being interconnected by a sliding connection along a translation axis orthogonal to the first and second rotational axes to form the clamping jaws of a clamp having a clamping axis parallel to the translation axis, the clamping axis being perpendicular to the first rotational axis and the second rotational axis of the gripping device.
2. The transportation unit according to claim 1, wherein the sliding connection comprises a first arm rigidly connected to the first jaw, a second arm rigidly connected to the second jaw, and a rod having a first end rigidly attached to either the first or the second arm and a second end mounted so that it can move in translation in a hole formed in the other of the first and the second arm.
3. The transportation unit according to claim 2, wherein the first arm and the second arm are identical.
4. The transportation unit according to claim 1, wherein at least one out of the first jaw and the second jaw comprises a centring V.
5. The transportation unit according to claim 1, wherein at least one out of the first jaw and the second jaw is flexible.
6. The transportation unit according to claim 1, wherein the first shuttle and the second shuttle are identical.
7. A conveyor comprising at least one of the transportation unit according to claim 1.